

0.1 High-dimensional experiments with QuadraSHAP

To assess whether Shapley attribution remains practical when the number of features greatly exceeds the number of patients, we consider two genomic survival datasets: GSE24080 multiple-myeloma gene expression [1] and TCGA lower-grade glioma (LGG) DNA methylation [2]. Genome-wide assays measure many molecular variables per patient, whereas the cohorts with matched survival follow-up contain only hundreds of patients. After eligibility filtering, GSE24080 contains 553 patients and 54,675 expression probe sets, and TCGA LGG contains 511 patients and 396,065 methylation sites. These are probe- or site-level features, retained without gene aggregation. We use the supplied 339/214 training/validation split for GSE24080 and a 383/128 split stratified by event status for TCGA LGG (seed 42), giving an extreme small- n , large- d setting.

We model overall survival using a Cox proportional-hazards model, which incorporates right-censored observations: patients whose death has not been observed by the end of follow-up. Its relative hazard factors across features, making QuadraSHAP applicable directly:

$$h(t | x) = h_0(t)f(x), \quad f(x) = \exp(\beta^\top x) = \prod_{j=1}^d \exp(\beta_j x_j).$$

To stabilize estimation when $d \gg n$, we shrink the Cox coefficients towards zero using a ridge penalty, with the implementation’s penalty parameter fixed a priori at $\alpha = 0.01d$. We use the Breslow approximation for patients with equal recorded event times. Missing values are imputed using training medians, and centering and scaling use training statistics. All retained features have nonzero fitted coefficients.

We explain the relative hazard using 30 training-background patients, sampled without replacement with seed 42 and held fixed across methods. Specifically, absent features are replaced by the corresponding entries of each background row and predictions are averaged:

$$v_x(S) = \frac{1}{30} \sum_{b=1}^{30} f(x_S, z_S^{(b)}).$$

The resulting game is a weighted sum of product games. We explain the first five held-out patients in each saved split with the degree-exact rule $m_q = \lceil d/2 \rceil$ and with patient-specific node counts selected to certify absolute quadrature tolerances $\varepsilon \in \{10^{-1}, 10^{-3}, 10^{-6}\}$. All methods use JAX-Metal FP32 on an Apple M4 Pro, an 8 GiB memory-planning budget, and one background row per block. We record one synchronized explanation call per patient and method after warming up each distinct dataset/node-count configuration. Reported times include automatic node selection, data transfers, and host accumulation, but exclude model fitting, warm-up, and cached quadrature-rule construction.

Table 1 shows that degree-exact computation remains manageable even at these dimensions: 27,338 and 198,033 nodes require an average of 1.17s and 66.38s per patient, respectively. For latency-sensitive use, the certified node budget reduces computation to 85–89ms for GSE24080 and 374–385ms for TCGA LGG. At $\varepsilon = 10^{-1}$, this corresponds to approximately 13.8 $\times$ and 177.2 $\times$ speedups, with all five patients in each cohort meeting the requested discrepancy threshold against the GPU degree-exact result. These subsecond latencies support interactive, on-demand Shapley explanations once the model and quadrature rules are ready. Caching matters: constructing the large exact rules previously required 10.97s and 561.92s on CPU, respectively, whereas approximate-rule construction took less than 1ms.

The numerical comparison also identifies the scope of the certificate. For patient p , let ϕ_p denote the exact Shapley vector of the fixed background game and $\phi_{m_p,p}$ its quadrature approximation in real arithmetic. The selected node count satisfies

$$\|\phi_{m_p,p} - \phi_p\|_\infty \leq C_p \leq \varepsilon, \quad C_{\max} = \max_{p=1,\dots,5} C_p.$$

The measured quantity instead compares floating-point outputs, $\Delta_{\text{GPU}} = \max_p \|\hat{\phi}_{m_p,p}^{\text{GPU}} - \hat{\phi}_{\text{exact},p}^{\text{GPU}}\|_\infty$. Although every selected rule satisfies its quadrature bound, the measured 10^{-3} threshold is met for only 4/5 GSE24080 patients and 2/5 TCGA LGG patients, and the 10^{-6} threshold is met for none. Moreover, GPU degree-exact outputs themselves differ from an independent, tightly converged FP64 reference by up to 1.87×10^{-3} and 3.32×10^{-2} , respectively, so agreement with that GPU reference does not establish exact numerical accuracy. These experiments therefore demonstrate rapid, quadrature-certified approximation, while stringent end-to-end GPU error guarantees require additional control of floating-point error. The certificate applies to the chosen 30-row background game and does not bound uncertainty from approximating the background distribution.

Dataset	Method	Nodes	GPU time (s)	$C_{\max}$	Δ_{GPU}	Pass
GSE24080	Exact degree	27,338	1.172 ± 0.035	0	—	—
	$\varepsilon = 10^{-1}$	44–53	0.085 ± 0.003	7.69×10^{-2}	3.40×10^{-4}	5/5
	$\varepsilon = 10^{-3}$	46–54	0.089 ± 0.010	8.82×10^{-4}	1.31×10^{-3}	4/5
	$\varepsilon = 10^{-6}$	48–57	0.086 ± 0.008	9.81×10^{-7}	8.59×10^{-4}	0/5
TCGA LGG	Exact degree	198,033	66.377 ± 1.196	0	—	—
	$\varepsilon = 10^{-1}$	37–49	0.375 ± 0.009	7.01×10^{-2}	4.47×10^{-3}	5/5
	$\varepsilon = 10^{-3}$	39–51	0.374 ± 0.019	5.80×10^{-4}	5.95×10^{-3}	2/5
	$\varepsilon = 10^{-6}$	41–53	0.385 ± 0.018	7.89×10^{-7}	1.07×10^{-2}	0/5

Table 1: Cox explanations on an Apple M4 Pro GPU using JAX-Metal FP32, 30 fixed training-background samples, and five distinct held-out patients. Time is mean $\pm$ sample standard deviation across patients, with one synchronized call per patient and method; node ranges span patients. $C_{\max}$ is the largest certified absolute quadrature bound across patients. Δ_{GPU} is the largest absolute difference from the matching GPU degree-exact result over all patients and features; Pass counts patients whose maximum feature-wise difference is at most ε . The certificate excludes floating-point error, and GPU degree-exact output is itself subject to rounding. Warm-up and quadrature-rule construction are excluded; automatic node selection is included. Previously measured CPU construction of the cached exact rules took 10.97 s (GSE24080) and 561.92 s (TCGA LGG); fresh approximate-rule construction took 0.418–0.625 ms (range of medians over five constructions).

References

- [1] NCBI Gene Expression Omnibus. GSE24080: MAQC-II project: Multiple myeloma data set. <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE24080>, 2010. Accessed 19 September 2026.
- [2] UCSC Xena. TCGA brain lower grade glioma (LGG) DNA methylation (HumanMethylation450). <https://xenabrowser.net/datapages/?dataset=TCGA.LGG.sampleMap%2FHumanMethylation450&host=https%3A%2F%2Ftcga.xenahubs.net>. Accessed 19 September 2026.